Percentage of carbon

44 g of CO_2 contain 12 g of C

0.245 g of CO_2 contains $\frac{12}{44} \times \frac{0.245}{0.26}$ g of C

% of Carbon = $\frac{12}{44} \times \frac{0.245}{0.26} \times 100 = 25.69$ %

Evaluate Yourself

5) 0.2346g of an organic compound containing C, H & O, on combustion gives 0.2754g of H_2O and 0.4488g CO_2. Calculate the % composition of C, H & O in the organic compound [C=52.17, H = 13.04, O = 34.79]

Estimation of sulphur:

Carius method: A known mass of the organic substance is heated strongly with fuming HNO_3. C & H get oxidized to CO_2 & H_2O while sulphur is oxidized to sulphuric acid as per the following reaction.

$$C \xrightarrow{\text{fum. } HNO_3} CO_2$$

$$2H \xrightarrow{\text{fum. } HNO_3} H_2O$$

$$S \longrightarrow SO_2 \xrightarrow{O + H_2O} H_2SO_4$$

The resulting solution is treated with excess of $BaCl_2$ solution H_2SO_4 present in the solution in quantitatively converted into $BaSO_4$, from the mass of $BaSO_4$, the mass of sulphur and hence the percentage of sulphur in the compound can be calculated.

Procedure:

A known mass of the organic compound is taken in clean carius tube and added a few mL of fuming HNO_3. The tube is the sealed. It is then placed in an iron tube and heated for about 5 hours. The tube is allowed to cool to room temperature and a small hole is made to allow gases produced inside to escape. The carius tube is broken and the content collected in a beaker. Excess of $BaCl_2$ is added to the beaker contains H_2SO_4 acid as a result of the reaction is converted to $BaSO_4$. The precipitate of $BaSO_4$ is filtered, washed, dried and weighed. From the mass of $BaSO_4$, percentage of S is found.

Mass of the organic compound = w g

Mass of the $BaSO_4$ formed = x g

233g of $BaSO_4$ contains 32 g of Sulphur

$\therefore$ x g of $BaSO_4$ contains $\left(\frac{32}{233} x\right)$ g of S

Percentage of Sulphur = $\left(\frac{32}{233} \times \frac{x}{w} \times 100\right)$ %

Example -2

In an estimation of sulphur by carius method, 0.2175 g of the substance gave 0.5825 g of $BaSO_4$ calculate the percentage composition of S in the compound.

Weight of organic compound 0.2175 g

Weight of $BaSO_4$ 0.5825 g

233 g of $BaSO_4$ contains 32 g of S

0.5825 g of $BaSO_4$ contains $\frac{32}{233} \times \frac{0.5825}{0.2175}$

Percentage of S=$\frac{32}{233} \times \frac{0.5825}{0.2175} \times 100$

= 36.78 %

Unit 11 revised - jagan.indd 142 22-01-2021 12:42:29